DOE Title Page

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

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Phase I

Primary Challenge and Focus Area: 21-A Artificial Intelligence in Fluid Flow for Energy Components and Technologies | Physics-Informed AI for Complex Flow Modeling

Senior/Key Personnel for Prime Applicant/Lead Institution and All Partner Institutions:
Florida State University (Lead): Mark Sussman, Kourosh Shoele
Sandia National Laboratories: Marco Arienti

Summary Budget Information for prime applicant/lead institutionand all partner institutions:
Budgets are Year 1 budgets
Florida State University (Lead) : 296096
Sandia National Laboratories: 150000
Total: 446096

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

1. Background/Introduction

Relevance to DOE Genesis Mission: This proposal responds directly to "The Genesis Mission: Transforming Science and Energy with AI," specifically Topic 21, "Artificial Intelligence in Fluid Flow for Energy Components and Technologies," and Focus Area 21A, "Physics-Informed AI for complex fluid phenomena."

We seek support to develop AI-enhanced numerical methods for atomization and spray processes that are central to combustion in aircraft engines [1] and rotating detonation engines used in rocket propulsion [2, 3]. These flows are especially challenging because they involve strongly coupled, multiscale interfacial dynamics that are difficult to resolve efficiently without losing predictive fidelity.

Our approach places AI within an efficient computational framework that uses high-fidelity simulations and targeted experimental data to train sub-scale models that generalize across atomization and spray problems. In direct alignment with Topic 21A, we will: **(1)** assimilate data from high-fidelity simulations and/or experiments into our existing multiphase-flow methodology [1–9]; and **(2)** use AI to learn dynamic, non-local corrections that recover the gap between under-resolved and high-fidelity simulations and/or experiments.

The overarching goal is to reduce the need for repeated high-fidelity simulations while maintaining physical consistency and predictive capability. More broadly, this work can strengthen predictive modeling for fuel injection, combustion, boiling, evaporation, condensation, and advanced propulsion systems, while advancing adaptive digital environments that tightly couple models, data, and computation.

Proposed Research in the context of the current state-of-the-art: Multiphase flows with *sharp, dynamically evolving interfaces* play a central role in energy conversion and transport systems, including fuel injection, spray combustion, rotating detonation engines, and phase-change-driven thermal processes [1–5, 10]. These flows are governed by *tightly coupled nonlinear interactions* among turbulence, interfacial forces, thermodynamics, and material response across a wide range of spatial and temporal scales, and are critical for identifying distinct mass and momentum processes at evolving interfaces. Their accurate prediction remains **one of the most challenging problems in computational science**.

Over the past several decades, semi-implicit **sharp-interface methods** have matured into a rigorous and reliable framework for modeling such systems. Foundational work by [11] established level set methods for tracking evolving interfaces in multiphase flow. Later advances produced conservative formulations [12] that significantly improved *mass conservation and robustness*, especially in the presence of topology change.

A major advance in this area is the **moment-of-fluid (MOF) [13–15]** and cell-integrated semi-Lagrangian **compressible MOF framework developed by Jemison, Sussman, and Arienti [16]**. The compressible multiphase formulation developed by Jemison et al. introduced a unified method that conserves mass, momentum, and energy while capturing shocks and material discontinuities without Riemann solvers. The material interfaces in [16] are represented sharply while preserving volume to machine precision in zero Mach number regions. Our approach enables simulations across compressible and incompressible fluids through asymptotic-preserving semi-implicit pressure updates. Further developments demonstrated accurate treatment of *multi-material junctions* and surface tension effects in complex geometries [2, 3, 5–8, 17]. However, the cost of resolving these multiscale interfacial processes at the fidelity required for predictive simulation remains high, making AI-enhanced modeling a natural and increasingly necessary next step.

This challenge is especially evident in high-fidelity DNS and LES studies of fuel injection and atomization, which reveal the *extreme computational cost* of resolving interface breakup, turbulence interaction, compressibility effects, and sharp mass transfer at fluid interfaces under realistic engine conditions. Such cost limits their usefulness for engineering design, where rapid turnaround over broad parameter spaces is essential. For liquid-jet-in-crossflow (JICF) atomization, even adaptive mesh refinement provides only limited practical benefit [10], because of the strong coupling between chaotic, bulk, gas-phase motion and liquid breakup. This motivates AI-enabled modeling and solution strategies that preserve predictive fidelity while overcoming fundamental computational bottlenecks.

One possible avenue of improvement we considered, as well as others [7, 18], was to replace costly geometric operations in multiphase flow algorithms with surrogate AI-generated models. The key principle is that surrogate models must not compromise on the fidelity and stability of the numerical algorithm. In particular, for volume tracking, or moment of fluid methods, it is essential that the "linearity preserving" and "volume preserving" properties of a multiphase flow system not be violated [13, 14, 19–21]. Yet, what was discovered by us [7], and also mentioned in [18] is that these AI generated surrogate algorithms, at best, provide a marginal benefit. We also bring attention to the following articles which give evidence that surrogate AI models will never beat finely tuned optimized algorithms [22–25].

Another possible direction is to pursue meshless methods such as PINN [26, 27] or SPH [28, 29]. However, these approaches, and pure meshless methods more generally, do not inherently preserve volume. This is a fundamental limitation, because in the low-Mach-number limit the fluid volume should remain conserved. Notably, the volume error reported for PINN in [26] was *worse* than that of classical level set methods. In this sense, they revisit the main criticism of early level set methods: lack of strict volume conservation.

It is proposed herein an alternative solution by pursuing two tightly connected advances: (I) we shall augment our existing compressible multiphase flow algorithm [2, 3, 16] with a volume- and mass-preserving compressible multiphase-flow nudging capability for the assimilation of data from high-resolution, high-fidelity simulations and/or experiments [30]. (II) We shall then use the nudging correction generated in (I) to train a conditioned AI model to construct sub-scale models for under-resolved multiphase flows. The resulting sub-scale model is expected to generalize across a broad class of multiphase flow problems.

2. Project Objectives

The project objectives are twofold:

OBJ. 1: We will advance our existing compressible multiphase flow algorithm [2,3,16] by developing an AI-centric compressible multiphase-flow nudging capability. The proposed nudging algorithm will enable volume- and mass-preserving "interface nudging" tailored to the complex atomization and spray problems considered here. Specifically, the nudging term will take the form $-\frac{p_s}{\rho}\nabla H$, where p_s is a nudging stress acting normal to the interface and is proportional to the difference between the simulated level set function and an assimilated level set function [30].

OBJ. 2: We will then use the nudging correction generated in (obj. 1) to train an AI model with regularization terms to derive sub-scale models for underresolved multiphase flows. The resulting model is expected to generalize across a broad range of multiphase flow regimes.

Our targeted canonical problems are: (a) atomization of a fuel jet in crossflow (JICF), which is directly relevant to combustion in aircraft engines [1]; and (b) the shock cylinder/drop problem, which is relevant to rotating detonation engines used in rocket propulsion [2, 3].

A successful Phase I outcome will create a clear pathway to **Phase II** by delivering AI foundations needed for extension to more realistic multiphase flow environments. This next step would target full three-dimensional, engine-relevant problems for predictive design and analysis.

3. Proposed Research and Methods; Physics-Constrained Data Driven Model

The proposed research builds on prior work by PIs **Sussman, Shoele, and Arienti** [5–8, 16], particularly the *cell-integrated semi-Lagrangian framework* and *moment-of-fluid (MOF) interface representation*, which enable accurate tracking of deforming interfaces with efficient treatment of compressibility and stiffness. The governing equations enforce conservation of mass, momentum, energy, and species, with extensions to phase change and related interface physics, including stress balance, mass-transfer-induced velocity jumps, and thermodynamic equilibrium.

Within this foundation, we propose a *physics-constrained, solver-embedded learning framework* in which high-fidelity computational and/or experimental data are assimilated, via nudging, into a coarse-grid two-phase-flow simulation, while neural networks learn unresolved interfacial and near-interface corrections from the resulting nudging source terms. Unlike PINN-style approaches that approximate the full solution field, this framework retains the conservative MOF-based solver, preserving sharp interface representation, strict mass conservation, and compatibility with established multiphase algorithms. AI therefore plays a focused role: learning localized corrections embedded directly into the governing equations rather than replacing the underlying physics-based solver.

The Phase I effort proceeds in three connected steps.

Task 1: we will implement a compressible multiphase nudging capability that assimilates interface, velocity, and temperature information from high-fidelity simulations and/or experiments into a coarse-grid solver while preserving volume and mass.

Task 2: we will develop the data-handling and restriction machinery needed to map fine-grid information onto the coarse-grid representation, thereby producing dynamically aligned coarse/high-fidelity state pairs.

Task 3: we will use the nudging correction generated by this alignment procedure to train a neural network that provides sub-scale corrections for underresolved multiphase flows. In this way, the AI model is trained not on arbitrary field mismatch, but on physics-based corrections extracted from an interface-resolved solver.

A central issue is that independently evolved coarse-grid and high-fidelity simulations diverge rapidly because interfacial errors amplify nonlinearly. As a result, the raw difference between the two solutions does not provide a clean learning target for unresolved physics. Instead, the coarse solution is periodically assimilated toward the restricted high-fidelity state, $Q^{n+1}_{hf\to c}$, and the aligned coarse state is then advanced by one coarse-grid time step. The discrepancy between this uncorrected update and the corresponding high-fidelity evolution defines the target correction that the neural network is trained to learn. This strategy ensures that the learned model captures missing interfacial and subgrid physics, rather than compensating for accumulated trajectory drift. Let Q^n_a denote the assimilated coarse state at time t_n, and let $Q^{n+1}_{c,unc} = G_{\mathrm{MOF}}(Q^n_a)$ denote the corresponding uncorrected coarse update. The target correction is

$$F^n_{\mathrm{corr}} = \frac{Q^{n+1}_{hf\to c} - Q^{n+1}_{c,unc}}{\Delta t}.$$

To implement the learned closure, we will construct training data directly from the nudging-based alignment procedure so that the neural network learns the one-step correction needed to recover high-fidelity interfacial dynamics from an underresolved coarse-grid update. The network inputs will include coarse-grid flow variables, thermodynamic fields, and reconstructed interface features—including volume fraction, centroids, interface normals, curvature, and a narrow-band interface mask—while the outputs will be localized correction terms added to the governing equations for momentum, energy, species, and, where needed, interface-normal dynamics. A neural network $\mathcal{N}_\theta$ will be trained to map coarse-grid flow and interface features, $z^n = \Phi(Q^n_a, \Gamma^n_a)$, to F^n_{corr} using a loss of the form $\mathcal{L}(\theta) = \sum_n \|W \cdot (\mathcal{N}_\theta(z^n) - F^n_{\mathrm{corr}})\|^2 + \mathcal{R}(\theta)$, where the weighting W and regularization $\mathcal{R}(\theta)$ are concentrated near the interface. As a practical Phase I step, we will use a localized residual/U-Net model for narrow-band correction learning and evaluate a neural-operator extension, *e.g.* U-FNO, to capture longer-range coupling and improve generalization across operating conditions.

Because the dominant coarse-grid errors arise in mixed cells and in a narrow band surrounding the reconstructed interface, both supervision and regularization are concentrated in these regions. The loss is weighted to emphasize interfacial discrepancies, while the regularization is designed to keep the learned correction localized, smooth, bounded, and geometrically consistent with the interface. In practice, this includes penalties that suppress spurious bulk corrections, reduce nonphysical oscillations, limit overly large predicted terms, and, for interfacial force corrections, promote alignment with the reconstructed interface normal. These constraints improve stability and keep the learned model focused on the unresolved interfacial physics most critical to predictive simulation.

When high-fidelity data are available, the nudging procedure aligns the coarse solution with reference dynamics at a fraction of the computational cost; when material interfaces are nudged, volume and mass preservation are enforced using the algorithm of Dommermuth et al. [30]. The resulting framework provides a clear and concise Phase I pathway: develop the assimilation capability, construct the learning target and neural-network correction model, and then verify and validate the integrated method on canonical atomization and spray problems. In this way, AI is used to extend the predictive reach of high-fidelity multiphase simulation while preserving the conservative structure and interface fidelity of the underlying solver.

4. Milestones and Deliverables

1. (Sussman, Months 1–3): Enhance the current compressible MOF algorithm [2, 3, 5, 16] by adding a nudging-based data-assimilation capability. Volume and mass conservation during nudging will be ensured using the algorithm of Dommermuth et al. [30]
2. (Arienti, Months 1–3): Develop routines to import high-resolution data [2, 3] into actively running coarse-grid simulations and construct the restriction operation $Q^{n+1}_{hf\to c}$.
3. (Shoele, Months 1–3): Develop the neural-network model that minimizes the discrepancy between predicted and target corrections, $\mathcal{L}(\theta) = \sum_n \|W \cdot (\mathcal{N}_\theta(z^n) - F^n_{\text{corr}})\|^2 + \mathcal{R}(\theta)$, with weighting W and regularization $\mathcal{R}(\theta)$ concentrated near material interfaces.
4. (Arienti, Sussman, Shoele, Months 4–6): Integrate Tasks 1–3, including nudging-based assimilation, AMReX data management, and AI-generated sub-scale models. Perform diagnostic testing, verification, and validation on 2D shock-cylinder atomization and spray problems.
5. (Arienti, Sussman, Shoele, Months 7–9): Extend Task 4 to fully 3D problems, with corresponding testing, verification, and validation.

5. Decision Gate Metrics

We will train the neural-network subscale models on the 2D/3D water and Galinstan shock-cylinder interaction problems shown in Figures 4, 6, and 7 of [3]. The effective fine-grid resolution for the high-fidelity simulation is $3072 \times 2304 \times 768$ (3D), while the corresponding low-fidelity resolution is $384 \times 288 \times 96$. After training, we will rerun the water and Galinstan cases using the learned model in place of direct high-fidelity assimilation and compare the resulting interfaces among: (a) coarse-grid assimilated results, (b) coarse-grid model-corrected results, and (c) high-fidelity results. We will then introduce two additional cases with varied Weber and Reynolds numbers to assess generalization. Interface errors will be quantified using the symmetric difference error [15].

6. Data Sources and Models

The current AMReX-based compressible multiphase solver used to generate Figure 3 of [1], Figures 4 and 6 of [3], and Figures 5 and 11a of [2] is maintained on GitHub at `https://github.com/msussman42/amrex_implicit_interfaces.git`. The data for Figure 3 of [1] and Figure 4 of [3] can be regenerated within 30 days. The remaining high-fidelity datasets are available through Marco Arienti on national laboratory computing resources.

7. Anticipated Phase I Outcomes and National Impact

Phase I will deliver an integrated AI-enhanced workflow for improved prediction of atomization and spray dynamics using limited high-fidelity simulation and/or experimental data. The key outcome will be a physics-constrained framework that combines compressible multiphase nudging, solver alignment, and learned sub-scale corrections within a conservative sharp-interface solver. This will lift the reliance on repeated high-fidelity calculations while preserving physical consistency and predictive fidelity. The broader impact is significant: improved predictive capability for atomization, spray breakup, and evaporation can strengthen computational design and analysis for energy-relevant multiphase systems. A successful Phase I will provide the foundation for Phase II, where the methodology will be extended to more realistic three-dimensional and operating conditions.

Appendix 1: Bibliography and References Cited
phase I
Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.
PI: Mark Sussman, Department of Mathematics, Florida State University
CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering
CO-PI: Marco Arienti, Sandia National Laboratories

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Appendix 2: Facilities and Other Resources
phase I
Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.
PI: Mark Sussman, Department of Mathematics, Florida State University
CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering
CO-PI: Marco Arienti, Sandia National Laboratories

Laboratory Resources: The FAMU-FSU College of Engineering performance site facilities used in this project are located within the innovation park at the joint college of engineering in Tallahassee. CO-PI Shoele is a member of Florida Center for Advanced Aero-Propulsion (FCAAP) and has a 270 square foot lab equipped with computers. Office space for graduate students are provided by FCAAP.

Computer (Shoele): CO-PI Shoele's laboratory is equipped with over a dozen PC workstations. Key software includes Tecplot, GridGen, Intel compilers. The department of Mechanical Engineering and FCAAP provide the software as well as printing facilities.

Computer (Sussman): PI Sussman has access in the math department to 8 4-core computers, 1 32 core machine, and 1 8-core machine. The 8-core machine has an Nvidea GPU card in it.

Computing Facilities (FSU-FAMU): There are a number of major computing facilities at FSU-FAMU that are available to the PIs Sussman, Shoele, and students in FSU-FAMU for this project. 1) HPC Cluster: 647 node, 10772 core, cluster with CUDA GPU capability attached to 256 TB Panasas file system that is mounted on all of the HPC compute nodes over a dedicated 1Gbps network. 2) Spear cluster: 88 x86 64-bit process cores linked together by a QDR Infiniband network. Eight of the nodes also connect to a PCI chassis equipped with 16 NVIDIA Tesla gpGPU cards (m2050) for a total 7,168 GPU cores for a peak system performance of 19.8 TFLOPS. 3) CAPS Real Time Digital Simulator (RTDS): A parallel-processing machine (performance at over 110 GigaFLOPS) with "14- rack" system consisting of 380 parallel processors and employing optimized EMTP-based nodal network simulation methods.

Computing Facilities (Sandia National Laboratories): Sandia National Laboratories' HPC Production Clusters for Institutional Programs provide computing resources through several high-bandwidth platforms (CPU- and GPU-based) optimized for scientific computing. In addition, Sandia offers state-of-the art file managing options for fast metadata operations and large data storage.

Appendix 3: Equipment
phase I
Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.
PI: Mark Sussman, Department of Mathematics, Florida State University
CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering
CO-PI: Marco Arienti, Sandia National Laboratories

The primary equipment that will be used for this project will be:

1. The FSU HPC Cluster will be used for generating high fidelity test data to be compared with the "low fidelity + AI generated model." Also, the FSU HPC Cluster will be used in the "AI model generation stage."

2. The math department 32 core and 8 core machines will be used for algorithmic development as well as the workstations in Shoele's lab.

3. At any given time, one of the Sandia National Laboratories' HPC Production Clusters will be available for the purpose of importing existing high fidelity data into running coarse-grid simulations.

Appendix 6: Transparency of Foreign Connections Disclosure and Certification phase I

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

PI: Mark Sussman, Department of Mathematics, Florida State University
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Appendix 7: Letters of Commitment from Sandia National Laboratories and Florida State University
phase I
Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.
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project summary

phase I

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PI: Mark Sussman, Department of Mathematics, Florida State University

CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering

CO-PI: Marco Arienti, Sandia National Laboratories

Fuel-spray atomization and evaporation sit at the heart of advanced combustion systems. In this proposal, we develop AI-enhanced numerical methods for “digital twin” atomization experiments designed to improve the efficiency and performance of jet engines and rotating detonation rocket engines.

What makes these problems so difficult is their multiscale character. Large liquid structures break into sheets and ligaments, then into droplets and mist, while evaporation continuously converts liquid fuel into a gaseous phase ready to react with oxidizer. Meanwhile, breakup, evaporation, and heat transfer reshape the surrounding turbulent flow, which then feeds back on the atomization process itself. In rotating detonation engines, the picture is even more delicate: some droplets survive the initial atomization stage and are later shattered by the detonation wave into a fine fog, after which reactions help drive the next cycle. The engine works best only within a narrow window, where the remaining pre-shock droplets are large enough to survive long enough to matter, but not so large that they reduce efficiency.

The difficulty, of course, is that the highest-fidelity simulations and the most informative experiments are both expensive. Our goal is therefore not to replace physics with AI, but to use AI to make better use of scarce high-value data within a conservative, interface-resolved simulation framework. The proposed work has three tightly connected parts: (1) data assimilation, (2) AI learning of multiphase-flow sub-scale corrections, and (3) verification and validation. In the first part, we will develop a compressible multiphase nudging capability that assimilates interface, velocity, and temperature information from high-fidelity simulations and/or experiments into a coarse-grid solver while preserving volume and mass. This step is essential because it provides a physically consistent way to align coarse-grid solutions with reference dynamics, rather than relying on raw coarse/fine mismatches that may be dominated by trajectory drift.

In the second part, the correction generated by this nudging-based alignment will be used to train a neural-network model to learn the unresolved interfacial and near-interface physics missing from underresolved simulations. In practice, the AI model will be trained on one-step correction terms extracted from an interface-resolved solver, using coarse-grid flow variables together with reconstructed interface features such as volume fraction, centroids, interface normals, and curvature. In this way, the learned model remains embedded within the governing equations of the underlying multiphase solver, where it acts as a localized, physics-informed correction rather than as a black-box surrogate. In the third part, we will test these learned corrections on atomization and spray problems outside the original training set, comparing coarse-grid results obtained with and without the AI model and quantifying the resulting improvement in interface prediction. The overall aim is to create a practical AI-enabled pathway for extending the predictive reach of high-fidelity multiphase simulation while preserving the physical structure and credibility of the underlying numerical method.